

Towards Improving the Robustness of Deepfake Audio Detection With Denoising Methods

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Abstract

Exploring the enhancement of deepfake audio detection robustness through the integration of denoising methods into traditional machine learning pipelines. Motivated by the increasing misuse of deepfake audio in phishing and identity spoofing, the study focuses on improving detection accuracy under real-world noisy conditions, particularly those introduced by VoIP transmission. To address this challenge, a denoising model based on Wave-U-Net is implemented to reconstruct clean audio from simulated VoIP-distorted samples. The ASVspoof2019 dataset is used to train and evaluate three baseline models: ResNet50 with LDA, LCNN, and XGBoost. Each model is assessed with and without the denoising layer to measure its impact on classification performance. Experimental results show that Wave-U-Net improves real audio detection across all models, with the most substantial gains observed in feature-based classifiers like XGBoost. LCNN demonstrates strong baseline robustness, while pipeline models benefit more from preprocessing. These findings confirm that denoising is a promising strategy for bridging the performance gap between clean and noisy audio environments, contributing to the development of more resilient deepfake audio detection systems.

CCS Concepts

• Security and privacy → Phishing.

Keywords

Deepfake Audio Detection, Wave-U-Net, Phishing, Denoise

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1 Introduction

Recent advancements in AI and machine learning have undeniably transformed entertainment and automation for the better. AI now plays a pivotal role in creating engaging social media content and enhancing customer service across industries. Over time, these technologies have become an integral part of everyday life, seamlessly integrating into human interaction and workflow. However, it is

also undeniable that AI has introduced negative consequences. Due to its ability to generate synthetic images, videos, and audio, AI technologies have been exploited to facilitate crimes and spread false information.

Cybercriminals are increasingly leveraging AI-driven techniques to develop malware, viruses, and sophisticated cyberattacks targeting organizations. Beyond technical threats, social engineering remains a crucial strategy, enabling attackers to manipulate individuals into providing sensitive information such as passwords or personal data. One of the most used and effective forms of cybercrime is phishing, where hackers deceive victims into providing confidential details or granting access to secure systems.

Deepfake technology has made phishing attacks even more dangerous, particularly through audio-based deception. Hackers can now impersonate trusted authority figures, such as a company's CEO, by generating highly convincing fake voices using advanced AI techniques like text-to-speech and voice conversion. For example, a hacker could create a deepfake voice to instruct an IT team to grant access to an account for which the hacker already possesses credentials.

As these threats continue to evolve, researchers and cybersecurity professionals are working urgently to develop robust detection systems capable of identifying deepfake audio and protecting individuals and organizations from AI-driven cyberattacks. Strengthening defenses against these emerging threats is essential to safeguarding digital security and maintaining trust in an AI-powered world.

To enhance detection accuracy, researchers have moved beyond traditional machine learning models and embraced deep learning approaches capable of identifying even the latest deepfake audio techniques. Various models have been developed to improve detection through attention mechanisms, transfer learning, adversarial learning, and other advanced techniques. As a result, the precision and reliability of deepfake audio detection continue to advance [19].

However, much of this research is conducted in controlled laboratory settings, focusing primarily on technical performance rather than practical application. This limited real-world implementation reduces the effectiveness of advanced detection models in identifying deepfake audio misuse, such as phishing scams and fraudulent calls.

To address the challenges of detecting deepfake audio under real-world noisy conditions, we focused on improving the robustness of detection models by introducing a denoising layer based on Wave-U-Net.



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- We implemented a **Wave-U-Net denoising model** to reconstruct clean audio from VoIP-distorted samples, simulating realistic phishing and identity spoofing scenarios.
- We evaluated the impact of denoising across **three baseline models** (ResNet50 with LDA, LCNN, and XGBoost) using the ASVspoof2019 dataset.
- We showed that **Wave-U-Net significantly improves detection of real audio samples**, with the most substantial gains observed in feature-based classifiers such as XGBoost.
- We demonstrated that **LCNN maintains strong baseline robustness** even without denoising, while pipeline models benefit more from preprocessing.
- We confirmed that denoising is a **promising strategy for bridging the gap** between clean laboratory datasets and noisy real-world environments, contributing to more resilient deepfake audio detection systems.

2 Related Works

2.1 Deepfake Audio Detection

Existing approaches to deepfake audio detection can be broadly categorized into three methodological paradigms: pipeline architectures, end-to-end deep learning frameworks, and feature-based machine learning classifiers.

Pipeline models, such as those built on ResNet50, emphasize interpretability and efficiency by combining feature extraction with dimensionality reduction before classification [3]. They benefit from transfer learning and structured workflows, but their reliance on spectrogram transformations makes them highly sensitive to noise and distortions.

End-to-end models, exemplified by LCNN, bypass handcrafted preprocessing and directly learn discriminative features from audio inputs [14]. This design offers strong robustness and adaptability, particularly in controlled settings, though it often comes with higher computational demands and challenges in real-time deployment.

In contrast, feature-based classifiers like XGBoost demonstrate the enduring relevance of traditional machine learning [2], leveraging handcrafted descriptors such as MFCCs and spectral features. These models are lightweight and interpretable, but their performance degrades sharply under noisy conditions, highlighting the importance of preprocessing and denoising.

2.2 Audio Preprocessing

For many deep learning models, such as CNN-based and traditional machine learning approaches for deepfake audio detection, the conversion of audio files into images is a necessary preprocessing step. This transformation allows the audio data to be fed into the model effectively. Typically, frequencies are represented on the y-axis, time is represented on the x-axis, and color intensity indicates amplitude.

However, as illustrated in Figure 1, most perceptually relevant audio content exists in the lower frequency range. When spectrograms are linearly scaled, a large portion of the image corresponds to high-frequency range where either silent or inaudible. Consequently, models are limited to learn from a part of the image input, which reduces their ability to extract meaningful patterns and features. To

enhance model performance and training efficiency, it is therefore essential to minimize the representation of silent or inaudible frequency range in the spectrogram. This adjustment ensures that the input image contains a higher density of informative, facilitating more effective learning.

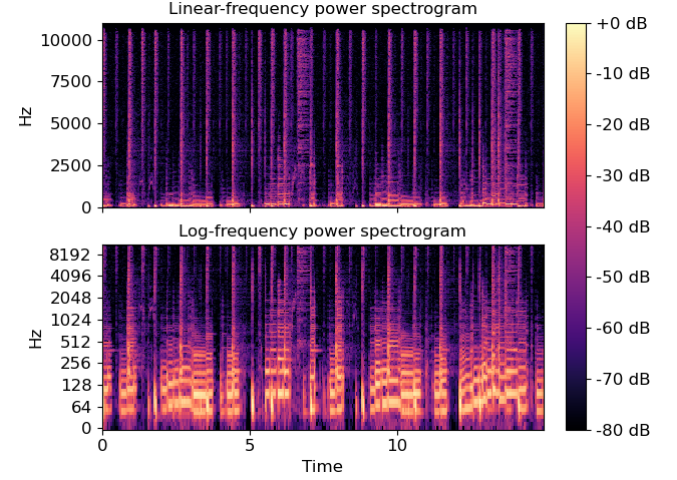


Figure 1: Linear-frequency Mel spectrogram (above) and log-frequency Mel spectrogram (below), showing differences in frequency scaling and perceptual emphasis [12]

A Mel spectrogram is a specialized type of spectrogram that accounts for the fact that human hearing is more sensitive to differences in lower frequencies. It scales the frequency axis (y-axis) accordingly, as illustrated in Figure 1 [12]. This transformation emphasizes perceptually relevant audio content, making it a preferred method for audio-to-image conversion among researchers [4, 5, 7–10, 16].

2.3 Denoising Methods

U-Net is an autoencoder-based architecture that uses a convolutional neural network (CNN) to compress input data via an encoder and subsequently reconstruct it using a decoder (Figure 2). Originally developed for medical image segmentation because of its ability to effectively link two images while maintaining high quality, U-Net’s application has expanded to image denoising and other image processing tasks. Because information is shared between layers, U-Net is known for producing high-quality images [15]. Despite being primarily an image processing model, its ability to denoise audio spectrograms is promising, making it a viable option for audio denoising tasks.

3 Proposed Approach

To enhance robustness against noisy audio input that better represents real-world conditions, such as phone calls and VoIP audio, we propose integrating a denoising layer into traditional models. This layer will clean the noisy audio before feeding it into the training model, effectively transforming the detection problem from a noisy environment into controlled lab settings, where many models proposed in the literature perform well.

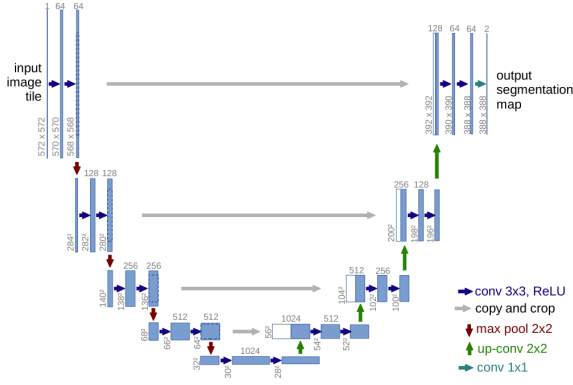


Figure 2: U-Net architecture, consisting of a contracting encoder and an expanding decoder [15]

Although some models’ robustness can be improved by simply augmenting the training set with additional noisy data [4, 18], my approach aims to transfer the performance of in-the-lab models to more realistic environments. By introducing denoising methods as a preprocessing step, this solution eliminates the need to increase the dataset size or retrain existing models while maintaining high detection accuracy.

This project focuses on improving the robustness of a traditional deepfake audio detection model when applied to real-world audio from a specific dataset. The goal is not to enhance the model’s generalization across diverse datasets, but rather to optimize its performance within the constraints of the selected data.

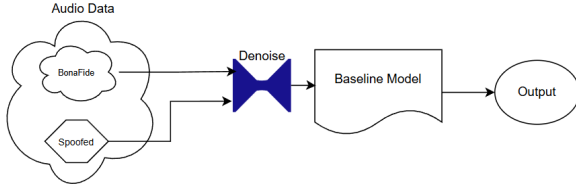


Figure 3: Proposed Framework. The denoising method is placed as the preprocessor of the training model

The framework of the proposed model integrates denoising models before the baseline models (Figure 3). This denoising section operates independently from the baseline models, requiring separate training procedures for its implementation.

3.1 Dataset

ASVspoof is one of the most widely used datasets for speaker verification spoofing detection and has gained significant traction in deepfake audio research [4–8]. Originally designed to enhance Automatic Speaker Verification (ASV) security, it provides high-quality genuine and spoofed speech data generated using advanced Text-to-Speech (TTS) and Voice Conversion (VC) methods, aiding researchers in training robust detection models [13, 17]. The dataset

is partitioned into Training, Development, and Evaluation subsets, with the Evaluation set containing deepfake techniques that are disjoint from those in the Training set. This design supports the evaluation of model generalization to unseen attack types.

3.2 VoIP Simulation

To train denoising models, paired datasets of clean and “noisy” audio are required. In this project, the “noisy” audio corresponds to real-world voice data commonly encountered in phone and voice calls. Such audio is typically transmitted via Voice over IP (VoIP) [1], which delivers digitized audio over the internet. Unlike traditional telephony, VoIP does not rely on dedicated service lines, making it a realistic source of noise for simulation and evaluation.

VoIP transmission can suffer from packet loss, latency, and compression, introducing distortions to the clean audio. These effects arise both from network conditions and from the codecs used to compress audio streams. In this project, the G.729 codec [1] is considered, as it is widely used and known to be lossy. According to the ITU-T standard, G.729 encodes audio at 8kHz, sufficient to capture the frequency range typical of human speech, focusing on frequencies below 4kHz.

3.3 Wave-U-Net

For the denoising task, we adapt U-Net—a convolutional neural network originally designed for biomedical image segmentation—to the audio domain. While the standard U-Net architecture is highly effective for image-to-image mapping, its application to audio typically requires converting signals into 2D spectrograms. We identified a significant limitation in this traditional approach: the transformations required for spectrogram-based denoising often result in the loss of phase information and the introduction of artifacts during the inverse Short-Time Fourier Transform (iSTFT) process. In the context of deepfake detection, these subtle signal distortions can inadvertently mask the very high-frequency artifacts that classifiers rely on to distinguish synthetic speech from human voices.

To address this, we specifically selected Wave-U-Net, a specialized variant designed to operate directly on raw audio waveforms in the time domain. By replacing standard convolutional layers with one-dimensional (1D) convolutions, Wave-U-Net enables an end-to-end mapping that preserves the original temporal structure of the audio. This architectural choice is technically justified because it ensures the denoised output remains in a raw waveform format, providing universal compatibility across our modular framework—specifically for models like LCNN that bypass handcrafted preprocessing to learn directly from raw inputs.

Furthermore, as demonstrated in [11], Wave-U-Net exhibits superior capabilities in signal inpainting and reconstructing missing frequency components. We leverage this property to combat the specific degradation caused by the G.729 codec. Since this codec introduces lossy compression by focusing on frequencies below 4kHz, Wave-U-Net is uniquely positioned to utilize contextual temporal information to restore the spectral fidelity lost during VoIP transmission. By restoring these components without the information loss associated with spectrogram transformations, we provide a more robust and high-fidelity input for the subsequent deepfake detection models.

4 Evaluation Method

To assess the effectiveness of denoising methods, three baseline models mentioned in the related works are chosen for evaluation. They are first evaluated on clean data (ASVspoof2019) and simulated data to establish initial performance metrics. Once the denoising method is trained, its impact is measured by comparing results against the same clean and noisy datasets.

Finally, the issue of class label balance in the training dataset. While the VoIP simulation process independently of the underlying class labels and the degradation mechanism itself does not favor any particular class, the reconstruction process performed by Wave-U-Net may still be influenced by class distribution. This project therefore compares models trained on original and balanced datasets to evaluate potential biases and generalisation effects. The evaluation process consists of the following steps:

Initial Performance: The baseline models are trained using ASVspoof2019 LA. Their training performance is assessed with a subset of ASVspoof2019 that was not used for training. Evaluation partition from ASVspoof2019 is used to measure the generalisation of the baseline model. Finally, the robustness of the baseline model is measured with the simulated VoIP audio.

Denoising Performance: After training both the unchanged Wave-U-Net and balanced Wave-U-Net, they are combined with the baseline model to see if the robustness against VoIP audio has improved or not. This is done on both the subset of Training partition, and Evaluation partition.

4.1 Evaluation Metrics

As deepfake audio detection is framed as a binary classification problem, we utilize standard evaluation metrics, including accuracy and the confusion matrix, to provide a comprehensive assessment of model performance.

However, in the context of phishing call detection, the cost associated with false positives is significantly higher than that of other errors as frequent false alarms can result in "alert fatigue" or the disruptive flagging of legitimate, high-priority communications.

Consequently, we place a particular emphasis on the False Positive Rate (FPR) and precision for real samples. This prioritization ensures that the evaluation aligns with cost-sensitive and user-centered security requirements, where minimizing the impact on legitimate users is as vital as the detection of spoofed content itself.

5 Results and Discussions

The experimental results demonstrate that incorporating Wave-U-Net denoising into the detection pipeline improves model robustness against noisy VoIP audio to some degree, particularly in recovering the detection of real audio samples. Across all three baseline models the addition of Wave-U-Net leads to improved recall for real samples, which are typically more affected by the simulated VoIP voices. However, the Wave-U-Net trained with class-balanced data, shows only marginal improvements over the unbalanced version, suggesting that the noise characteristics may play a more dominant role than class distribution in influencing detection performance.

For the ResNet50 model (Table 1), which relies on a pipeline architecture with LDA, the results indicate a strong sensitivity to noise. Without denoising, the model performs poorly on real audio,

Table 1: ResNet50-Precision/Recall Result from the Experiment

ResNet50				
	Train		VoIP	
	precision	recall	precision	recall
Fake	0.99	1.00	0.91	0.99
Real	0.96	0.95	0.56	0.10
Wave-U-Net		Balanced Wave-U-Net		
	precision	recall	precision	recall
Fake	0.95	0.95	0.95	0.95
Real	0.55	0.52	0.56	0.58
Evaluation		VoIP		
	precision	recall	precision	recall
Fake	0.99	0.87	0.91	0.99
Real	0.46	0.92	0.69	0.11
Wave-U-Net		Balanced Wave-U-Net		
	precision	recall	precision	recall
Fake	0.94	0.92	0.94	0.91
Real	0.39	0.45	0.40	0.51

Table 2: LCNN-Precision/Recall Result from the Experiment

LCNN				
	Train		VoIP	
	precision	recall	precision	recall
Fake	1.00	1.00	0.99	0.96
Real	0.98	0.97	0.73	0.94
Wave-U-Net		Balanced Wave-U-Net		
	precision	recall	precision	recall
Fake	0.99	0.97	0.99	0.97
Real	0.76	0.92	0.90	0.97
Evaluation		VoIP		
	precision	recall	precision	recall
Fake	0.98	0.90	0.98	0.84
Real	0.47	0.81	0.38	0.83
Wave-U-Net		Balanced Wave-U-Net		
	precision	recall	precision	recall
Fake	0.97	0.84	0.97	0.86
Real	0.36	0.78	0.39	0.76

with recall values of 0.10–0.11. However, when Wave-U-Net is introduced, real recall improves to around 0.52–0.58. These findings suggest that while ResNet50 benefits from denoising, it may still struggle to generalise across both classes under noisy conditions.

In contrast, the LCNN model (Table 2), which is trained end-to-end, shows strong performance even without denoising. It achieves high fake recall and high real recall (around 0.97) in the training phase. The addition of Wave-U-Net slightly reduces fake recall

Table 3: XGBoost-Precision/Recall Result from the Experiment

XGBoost					
	Train		VoIP		
	precision	recall	precision	recall	
Fake	0.99	0.99	0.90	1.00	
Real	0.95	0.87	0.77	0.03	
	Wave-U-Net		Balanced Wave-U-Net		
	precision	recall	precision	recall	
Fake	0.96	0.90	0.95	0.92	
Real	0.42	0.66	0.46	0.58	
	Evaluation		VoIP		
	precision	recall	precision	recall	
Fake	0.97	0.91	0.90	1.00	
Real	0.47	0.74	0.68	0.04	
	Wave-U-Net		Balanced Wave-U-Net		
	precision	recall	precision	recall	
Fake	0.95	0.87	0.94	0.89	
Real	0.34	0.59	0.35	0.53	

but maintains high real recall (0.76–0.83), indicating that LCNN is inherently more robust to noise.

The XGBoost model (Table 3), which operates on extracted features, shows the most dramatic improvement from denoising. Without Wave-U-Net, its real recall is 0.02–0.04, but with denoising, it rises to 0.53–0.59. This suggests that XGBoost is highly susceptible to noise in raw features and benefits significantly from pre-processing. However, even with denoising, its performance remains below that of the deep learning models, highlighting its limitations in handling complex audio distortions.

Overall, the results reveal a consistent pattern. Fake audio detection remains robust across all models and conditions, while real audio detection is the primary challenge under noisy VoIP scenarios. Wave-U-Net effectively mitigates this issue, especially for models like ResNet50 and XGBoost. The Balanced Wave-U-Net variant offers slight improvements in real recall but does not consistently outperform the unbalanced version, indicating that future work may need to explore alternative strategies beyond class balancing to further enhance real audio detection.

6 Limitation

Despite the promising improvements introduced by Wave-U-Net denoising, there are several limitations around generalisation and practical deployment of the proposed framework.

First, the denoising model was trained using simulated VoIP audio generated via SoX with the G.729 codec. While this codec is widely used and representative of real-world compression artifacts, the simulation does not fully capture the diversity of distortions observed in real VoIP environments. Real-world audio may include packet loss, jitter, and echo that are not present in the simulated

audio. This gap between simulated and authentic noisy conditions may limit the robustness of the denoising model.

Another key limitation lies in the dataset selection. This project uses ASVspoof2019, which contains clean, studio-recorded audio. While this choice ensures high-quality training data and facilitates controlled evaluation, the absence of real-world audio samples, such as those found in the In-the-Wild or Fake-or-Real datasets, limits the model’s generalisability to diverse acoustic environments. Moreover, relying on a one dataset may introduce bias in the training, especially if the dataset lacks variability in language, accents, or emotional tone.

One practical limitation of this project is the absence of an evaluation on the minimum computational resources required to deploy the model in real-time. This consideration is particularly important in the context of phishing call detection, where low-latency response and deployment on resource-constrained devices such as mobile phones are often necessary. While models like LCNN demonstrate strong performance, their computational complexity may render them less suitable for real-time applications if the computational cost is too high.

Interestingly, the experimental results suggest that the frequency components removed during VoIP simulation have minimal impact on LCNN’s performance. This observation implies that the most discriminative features for distinguishing deepfake audio from genuine human speech may reside predominantly in the lower, more perceptible frequency bands. Consequently, the loss function used to train the Wave-U-Net should prioritise reconstruction of these lower-frequency components, effectively preserving the features most critical for classification.

A subtle yet important implementation detail in the current Wave-U-Net setup is the fixed input length of 4 seconds. This duration was chosen to ensure compatibility across all three baseline models and is achieved by trimming or repeating the input audio as needed. However, this approach introduces a potential mismatch with models like LCNN, which handle short audio samples using zero-padding rather than repetition. Repeating audio to meet a fixed length may result in denoised outputs that deviate from the distribution seen during training, effectively creating out-of-distribution inputs for certain models. To address this, a more robust strategy would involve training Wave-U-Net on fixed-length audio windows and reconstructing full-length outputs by concatenating the denoised segments. This approach would preserve the original temporal structure of the input audio and improve compatibility.

7 Future Work

To advance the robustness and real-world applicability of deepfake audio detection, several future works are suggested. While Wave-U-Net denoising has shown improvements in handling simulated VoIP distortions, future work should focus on bridging the gap between simulated and authentic audio environments.

The next step is to integrate Twilio’s programmable Voice API into the framework as utilised by [4]. Twilio enables the injection of audio samples into real VoIP call flows, allowing researchers to capture audio that has traveled actual network infrastructure. This includes exposure to codec negotiation, jitter, latency, and device-specific artifacts that are difficult to replicate using offline

tools. By routing both clean and spoofed samples through Twilio and recording the returned audio, future experiments can measure detection performance under realistic conditions.

Alternative denoising models should also be explored and investigated. There may be other models that offer improved generalisation and better preservation of speech features critical to spoof detection.

From a deployment perspective, optimising the pipeline for real-time performance is essential. Techniques such as model pruning, quantisation, and knowledge distillation could reduce inference latency, making the system viable for live call monitoring.

Finally, expanding the evaluation framework to include cost-sensitive and user-centered metrics will provide a more nuanced understanding of model behavior. Metrics such as ROC-AUC, F1-score, and false positive cost weighting are critical in security contexts where detection errors have operational consequences. Incorporating explainability tools like SHAP or LIME could further enhance analyst trust and facilitate human-in-the-loop verification.

8 Conclusion

This project investigated the enhancement of deepfake audio detection robustness through the denoising methods, specifically Wave-U-Net. The current detection models lack the evaluation under real-world noisy conditions and this study addressed the gap between controlled lab performance and practical deployment. Drawing from literature on pipeline and end-to-end architectures, as well as audio preprocessing techniques such as mel-spectrogram conversion, the project implemented and evaluated three baseline models: ResNet50 with LDA, LCNN, and XGBoost.

Experimental results confirmed that denoising improves real audio detection under simulated VoIP conditions to some degree, with Wave-U-Net enhancing recall across all models. The LCNN model demonstrated inherent robustness, while XGBoost benefited most from denoising. Balanced training of the denoising model achieved only marginal improvement, suggesting that noise characteristics outweigh class distribution in influencing detection accuracy.

To continue and improve this work, future research should incorporate real-world noisy datasets to test generalisation beyond synthetic distortions, enabling valid benchmarking and real-time deployment.

This study lays the groundwork for resilient deepfake audio detection systems capable of operating in real-world conditions. By aligning technical innovation with practical constraints, it contributes to the goal of safeguarding digital trust from increasingly sophisticated synthetic media.

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